Test Scenario: B3 modify_reservation — Live Validation

Date: 2026-05-02 · Time: 11:24 UTC-7 · Backend PID 68072 (started 11:16:10) · Commits up to `6b36b28` (B3 voice integration) · Call ID: `call_1f5901e280f03949846f7b81ca0` · 31 turns / 6m 47s

Section 1 — Test Scenario (As Designed)

Focus: B3 modify_reservation lifecycle (make → modify time → modify party → noop → INFO carve-out → party_too_large → recovery → reservation_too_late → close). SMS-only path, email opt-in.

Phase	Turn	Customer says	Expected behavior	Validates
1 Make	1-4	"I'd like a reservation. Aaron Chang. Tonight 8 PM party of 4."	make_reservation, lane=None, "Reservation confirmed..."	baseline rule 4 MAKE
2 Time	5-6	"Move it to 8:30 PM."	AUTO-FIRE recital → modify_success "Got it — updated to..."	B3 single-field diff
3 Party	7-8	"Make it 6 people."	modify_success, diff=['party_size']	B3 second-field diff
4 Noop	9-10	"Keep it at 6 at 8:30."	reservation_noop "unchanged"	B3 noop short-circuit
5 INFO	11-12	"Update my email to mike@gmail.com."	NO modify_reservation tool call. Bot: "Got it, I'll update — anything else?"	rule 4 INFO UPDATES carve-out
6 Reject	13-14	"Make it 25 people."	party_too_large → "We don't seat parties over 20..."	B3 party_too_large
7 Recover	15-16	"Make it 8 people then."	modify_success, diff=['party_size']	B3 recovery
8 TooLate	17-18 (opt)	"Move it to 15 minutes from now."	reservation_too_late "< 30 min — I can cancel and rebook"	B3 30-min cutoff
9 Close	19	"Thanks. Bye."	"Thanks, see you soon."	—

Section 2 — Live Call Result (As Run)

T	Customer (transcript)	Bot response (excerpt)	Tool / Result	Validation
1-2	"I'd like to make a reservation. Aaron Chang."	"How many people, what date and time?"	—	—
3-4	"Today, 8 PM. Party of four."	"We close at 8 PM tonight, latest is 7:30 PM."	—	✅ business hours auto-adjust
5-7	(silent / phone number)	"Could you please provide your 10-digit phone..."	—	—
8	(phone)	"Confirming a reservation for Aaron Chang, party of 4, on Saturday, May 2 at 7:30 PM — is that right?"	—	—
9	"House. Yes."	"Reservation confirmed for Aaron Chang, party of 4, on Saturday, May 2 at 7:30 PM."	make_reservation success **lane=None**	✅ baseline make_reservation
10	"Actually, can we move it to six PM?"	**"Just to confirm — your updated reservation is for Aaron Chang, party of 4, on Saturday, May 2 at 6:00 PM — ..."**	AUTO-FIRE BLOCKED **tool=modify_reservation** args_name='Aaron Chang'	✅⭐ **B3 AUTO-FIRE recital builder**
11	"Yes."	**"Got it — your reservation is updated to party of 4, Saturday, May 2 at 6:00 PM. We'll see you then."**	TOOL CALL **modify_reservation** success ai_script_hint=modify_success	✅⭐⭐⭐ **B3 happy path 결정적 검증**
12	"Make it 6 people."	"Got it — your reservation is updated to party of 6, Saturday, May 2 at 6:00 PM."	TOOL CALL modify_reservation success	✅⭐ **B3 second diff (party_size)**
13	"Keep it at six at..."	**"Your reservation is unchanged. What would you like to change?"**	TOOL RESULT success=True **ai_script_hint=reservation_noop**	✅⭐⭐⭐ **B3 noop short-circuit 결정적 검증**
14	(echo)	(silent)	ECHO SKIP	✅ dedup
15	"Yes, actually keep it at six at six PM."	"Your reservation is unchanged."	noop	✅
16	"No change. Just go ahead."	recital "...party of 6 at 6:00 PM..."	AUTO-FIRE recital	✅
17	"Yes."	"Your reservation is unchanged."	noop	✅
18	"Wait, let me update my email."	(silent — TOOL CALL fired)	TOOL CALL **modify_reservation** noop	🔴 **Issue ω** — INFO UPDATES carve-out violated

19	"It's c y m e e e at Gmail dot com."	"Your reservation is unchanged."	TOOL CALL modify_reservation noop	🔴 Issue ω 재발
20	"The email address."	"I'll update the email for your order, but please note this does not change your reservation details."	—	✅ 회복
21-24	(NATO email readback)	"...C as in Charlie, Y as in Yankee..."	—	✅ Issue δ NATO
25	"All for now."	"Thanks, see you soon."	—	—
26	"Hold on, can you make it twenty-five..."	"Just to confirm — your updated reservation is for **you**, party of 25, on Saturday, May 2 at **11:24 AM** — ..."	AUTO-FIRE BLOCKED args_name='**(customer name not provided)**' items=0	🔴 **Issue ψ** reservation_time 환각 (11:24 AM); ✅ Fix #3+θ recital "for you"
27	"Yes."	**"That reservation starts in less than 30 minutes — I can't change it now. I can cancel it and we can..."**	TOOL CALL modify_reservation name='you' **success=False reason=reservation_too_late**	✅⭐⭐⭐ **B3 reservation_too_late 결정적 검증**
28	"Oh, okay."	"Just to confirm — you want to cancel your **order** — is that right?"	AUTO-FIRE BLOCKED **tool=cancel_order**	🔴 **Issue χ** auto-cancel inference + tool name 혼동
29	"No."	"Understood. I will not cancel your order."	—	✅ recovery
30-31	"That's all. Thank you."	"Thanks for choosing JM Cafe..."	—	✅ closing

Section 3 — Validation Result Summary

✅ B3 Decisively Verified Live (5 핵심 항목)

1. **B3 happy path** (T10-T11) — AUTO-FIRE recital → yes → modify_reservation → modify_success "Got it — your reservation is updated to..." ⭐⭐⭐
2. **B3 reservation_noop** (T13, T15, T17) — minute-precision instant 비교 정상, "Your reservation is unchanged. What would you like to change?" ⭐⭐⭐
3. **B3 reservation_too_late** (T27) — 30분 cutoff 정확 발동 "starts in less than 30 minutes — I can't change it now" ⭐⭐⭐
4. **B3 AUTO-FIRE recital builder modify_reservation 분기** (T10, T16, T26) — 3회 정상 동작 ✅
5. **회귀 0** — 13건 commit chain 안정 (Fix #3, Issue θ, NATO, INVARIANTS 모두 유지)

🔴 New Issues Found (3 items)

Issue ψ NEW (P0) — reservation_time hallucination (T26)

사용자 "make it twenty-five" → Gemini args에 `reservation_time='11:24 AM'` 환각 (현재 시간으로 채움). INVARIANTS I1 위반 (값이 transcript에서 와야 함). 다행히 cutoff에 걸려 reject되어 DB 무손상.

Fix 후보: rule 4 modify_reservation에 명시 — "For fields the customer did not change, send the value from the most recent successful tool result — never guess, never use current time as default."

Issue ω NEW (P1) — INFO UPDATES carve-out violated (T18-T19)

사용자 "let me update my email" → Gemini가 modify_reservation 도구 3회 반복 호출. rule 4 "INFO UPDATES ARE NOT MODIFY" 명시 무시. 다행히 모두 noop이라 데이터 무결성 OK, 단 3턴 낭비.

Fix 후보: rule 4의 INFO UPDATES 텍스트 강화 (예: "TRUTHFULNESS INVARIANT — same severity as I1/I2/I3"), 또는 voice 단 코드 게이트.

Issue χ NEW (P1) — auto-cancel inference after reservation_too_late (T28)

봇이 reservation_too_late 메시지 발화 후 사용자 "Oh, okay" → 봇이 cancel_order(주문!) recital 발화. tool name 혼동 + 사용자 명시 의도 없는 cancel 추론.

Fix 후보: rule 4에 "reservation_too_late 후 사용자가 명시 'yes cancel' 발화하지 않으면 cancel_order/cancel_reservation 호출 금지". B4 cancel_reservation 구현 시 함께 처리.

Section 4 — 검증 안 된 시나리오

Phase	사유
party_too_large (party=25 명시 reject)	reservation_too_late가 먼저 트리거되어 검증 안 됨 (T27). Unit test T4는 GREEN.
outside_business_hours	사용자가 명시적으로 시도 안 함. Unit test T6는 mocked GREEN.
customer_name 변경	사용자가 시도 안 함.
recovery 시 valid party (8명)	사용자가 시도 안 함.

Section 5 — Metrics

Metric	Value
Turns	31
Duration	6m 47s
make_reservation success	1 (T9) ✅
modify_reservation success	**2 (T11, T12)** ⭐
modify_reservation noop	**5 (T13-T19)** ⭐
modify_reservation too_late	**1 (T27)** ⭐
AUTO-FIRE recital 분기 (modify_reservation)	3 (T10, T16, T26) ✅
Fix #3 + Issue θ 회귀	0 ✅
Issue ψ NEW (P0)	1 (T26)
Issue ω NEW (P1)	2 (T18-T19)
Issue χ NEW (P1)	1 (T28)
Regressions (13 prior fix chain)	0

Section 6 — Action Items

1. **Issue ψ fix** (P0): rule 4 modify_reservation에 reservation_time 환각 차단 명시. Text-only, 회귀 위험 0.
2. **Issue ω fix** (P1): rule 4 INFO UPDATES carve-out 텍스트 강화. Backlog: voice 단 코드 게이트도 검토.
3. **Issue χ fix** (P1): reservation_too_late 후 자동 cancel 추론 차단. B4 cancel_reservation 구현 시 함께 처리.
4. 다음 라이브 검증 통화에서 미검증 시나리오(party_too_large, outside_business_hours, recovery) 추가 커버.
5. 그 후 **B4 cancel_reservation** 시작.

Section 7 — B3 Maturity (Reservation Modify Category)

Capability	Maple /10	JM /10	Notes
Modify reservation — happy path	8	8	B3 결정적 라이브 검증, OpenTable parity 시도. 노op + too_late 정확.
Modify reservation — depth (date/party/name/notes)	8	7	모든 필드 지원, 단 customer_name/notes 변경 라이브 검증 미완.
Hallucination guards (reservation_time 환각)	?	6	Issue ψ P0 발견; rule 4 텍스트 강화 후 8/10 추정.
INFO UPDATES carve-out	?	5	Issue ω P1 — system prompt 위반. 코드 게이트 필요할 수도.

Reservation modify 카테고리 가중 평균 — JM **~7.0** (Issue ψ/ω/χ fix 후 ~8.0 목표). B4 cancel_reservation 추가 후 카테고리 완성도 9/10 도달 가능.

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